DATA COLLECTION REPORT

PlaySafe

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1.Introduction

This report describes the data collection and initial inspection process for the PlaySafe project. The aim of the project is to investigate whether information about LaLiga players, their playing context and previous injury history can be used to predict whether a player will experience an injury in a subsequent season.

The report explains how the dataset was selected, what information it contains and what was identified during the initial inspection of the data. It also compares the available information with the data requirements defined in the project proposal. Finally, the report identifies the main data-related issues that need to be considered before continuing with data preparation, exploratory analysis and modelling.

2. Dataset selection and overview

I started the project by searching for a suitable dataset for the injury prediction problem.

The dataset used in this project was obtained from the GitHub repository football-injury-scraper. The specific file used is laliga_players.csv.

laliga_players.csv – football-injury-scraper repository

I selected this dataset because it contains information about LaLiga players, their seasons, injuries, playing time and other player characteristics. These variables provide a suitable starting point for investigating whether player information and previous injury-related information can be used to predict injuries in a subsequent season.

After obtaining the dataset, I loaded it into the analysis environment and reviewed its structure before starting the data cleaning process.

3.Initial data inspection

The dataset contains 3,713 records and 15 variables. I checked the structure of the dataset, the data types, missing values and the format of the different variables.

Several data quality issues were identified. All variables were initially stored as strings, including numerical variables and dates. Some variables also contained values with additional text, such as "187 m" for height and "32 days" for the duration of an injury.

I identified missing values in several columns. In particular, the injury-related variables contain a large number of empty values. I also found 96 "-" values in matches_missed.

Two age variables, current_age_x and current_age_y, appear to represent the same concept but use different formats. current_age_x contains no empty values, while current_age_y contains missing values.

I also identified possible character encoding issues in some player names, such as "GalÃ¡n".

An important observation was that missing injury information should not automatically be interpreted as meaning that a player was never injured. This needs to be considered when defining the target variable for the prediction model.

Outcome

The initial inspection provided an overview of the dataset and identified the main data quality issues that need to be addressed. The next step is to clean and standardise the data, convert variables to appropriate data types and investigate the identified issues before continuing with exploratory analysis and feature analysis.

4.Data Requirements check:

The purpose if this point is to show whether the dataset contains the information, I classified as necessary in my Proposal.

Requirement	Available
LaLiga players	Yes
Season information	Yes
Playing context	Yes
Injury information	Yes
Player profile	Yes
Variable has_injury_next _season	Not directly available

Almost all variables are available in the dataset which will be very helpful for developing the model. However, the target variable is not currently present in the dataset and needs to be investigated, The available season and injury information may allow this target to be constructed.

5. Conclusion:

The initial data collection and inspection showed that the selected dataset contains several variables that are relevant to the research question, including player characteristics, season information, playing time and injury-related information. The dataset therefore provides a suitable starting point for the project.

I also identified several data quality issues, including inconsistent data types and formats, missing values, encoding issues and potentially redundant variables. An important point that still needs to be investigated is the definition of the target variable, has_injury_next_season. This variable is not directly available in the dataset and therefore needs to be investigated before modelling